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Optimization: A Journal of Mathematical Programming and Operations Research
Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/gopt20

A special newton-type optimization method

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Version of record first published: 20 Mar 2007.

To cite this article: A. Fischer (1992): A special newton-type optimization method, Optimization: A Journal of Mathematical Programming and Operations Research, 24:3-4, 269-284

To link to this article: http://dx.doi.org/10.1080/02331939208843795

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Optimization, 1992, Vol. 24, pp. 269–284
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Printed in the United Kingdom

A SPECIAL NEWTON-TYPE OPTIMIZATION METHOD

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(*Received 29 October 1990; in final form 8 November 1991*)

The Kuhn–Tucker conditions of an optimization problem with inequality constraints are transformed equivalently into a special nonlinear system of equations $T_0(z) = 0$. It is shown that Newton's method for solving this system combines two valuable properties: The local Q-quadratic convergence without assuming the strict complementary slackness condition and the regularity of the Jacobian of T_0 at a point z under reasonable conditions, so that Newton's method can be used also far from a Kuhn–Tucker point.

KEY WORDS Nonlinear programming algorithms, inequality constraints, Kuhn–Tucker points, Newton's method, Clarke's Jacobian, strict complementary slackness condition

Mathematics Subject Classification 1991:
Primary: 65K05, Secondary: 90C30

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1. INTRODUCTION

The nonlinear programming problem to be considered in this paper is given by

$$f(x) = \min! \quad \text{subject to} \quad g^i(x) \leqq 0 \qquad i \in I := \{1, \ldots, m\}, \tag{1.1}$$

where $f, g^1, \ldots, g^m$ are twice continuously differentiable functionals defined on $\boldsymbol{R}^n$. Extensions to problems including also equality constraints will be obvious.

For solving such problems so-called Newton-type Optimization Methods (NOM) have been developed in the last years ([8, 11, 12, 13, 20, 26]). The basic idea of these methods (cf. [1, 6, 14, 30]) consists in transforming the system of Kuhn–Tucker conditions

$$\begin{gathered} \nabla f(x) + \sum_{i=1}^{m} u^i \nabla g^i(x) = 0 \\ u^i \geqq 0,\ g^i(x) \leqq 0, \qquad u^i g^i(x) = 0 \qquad (i \in I) \end{gathered} \tag{1.2}$$

into an equivalent nonlinear equation

$$T(z) = 0. \tag{1.3}$$

The mapping $T: \boldsymbol{R}^p \to \boldsymbol{R}^p$ is defined such that each solution $z^* \in \boldsymbol{R}^p$ of (1.3) can be assigned to a solution $(x^*, u^*) \in \boldsymbol{R}^n \times \boldsymbol{R}^m$ of (1.2) and vice versa. Then, the equation (1.3) is solved iteratively using a Newton-type method. Along these lines, by choosing both the mapping T and the Newton-type method various locally superlinearly convergent NOM for computing a Kuhn–Tucker point (x^*, u^*) of (1.1) are possible.

Over against numerous SQP-methods (e.g. [22, 25, 31]) the NOM permit that the subproblems occurring in each iteration are linear equations instead of quadratic programs with linear inequality constraints. Furthermore, a strategy for the precise estimation of the active index set $I(x^*)$, as used by various optimization algorithms (e.g. [4, 10, 19]), is not necessary for Newton-type Optimization Methods.

From experience, Newton-type methods for solving equations, possibly coupled with a procedure for enforcing global convergence, frequently work successfully also far from a solution. Under that global point of view, the existence of the inverse $\nabla T(z)^{-1}$ is an essential question for NOM.

In this paper we suggest a Newton-type Optimization Method based on a special mapping T_0. In contrast with the existing NOM known to the author, the new method combines two valuable properties:

(i) The local quadratic rate of convergence can be proved *without* assuming the strict complementary slackness condition

$$I(x^*) = I^+(x^*, u^*), \tag{1.4}$$

where $I(x) := \{i \in I \mid g^i(x) = 0\}$ and $I^+(x, u) := \{i \in I(x) \mid u^i > 0\}$.

(ii) The Jacobian of T at a point $z \in \boldsymbol{R}^p$ is regular under reasonable conditions, in particular concerning the question, for which indices $i \in I$ the linear independence of the vectors $\nabla g^i(x)$ must be required.

All NOM developed up to now possess one of these properties at most, (except the brief previous report of Fischer [8]).

The nonsmooth mapping T_0 is considered in Section 2. Before stating Algorithm $\mathscr{A}$ in Section 3 extensions of Newton's method for nonsmooth mappings are briefly discussed. In particular, the investigations by Kojima/Shindo [15] and, more general, of Kummer [16, 17] turn out to be of special importance for our aim that the subproblems of Algorithm $\mathscr{A}$ are *linear equations*. In Section 4 the algorithm is studied with respect to (i) and (ii). As an illustration of the behaviour of existing NOM related to these two properties three examples are analyzed in Section 5. Computational results follow at the end of the paper.

The symbols ∇ and ∇^2 denote the gradient and the Hessian of a functional, in case of differentiation with respect to x we use ∇_x and ∇^2_{xx} accordingly. The Jacobian of a mapping $F: \boldsymbol{R}^p \to \boldsymbol{R}^p$ is written as ∇F^T. Moreover, the subscript k denotes the iteration index, whereas the superscript i refers to the index of the constraint function g^i. For abbreviation we set $a_+ := \max(0, a)$ for any real number a and $w_+ := (\cdots(w^i)_+\cdots)^T$ for any real vector w. Furthermore, the vector $g(x) := (g^1(x), \ldots, g^m(x))^T$ is introduced.

2. THE MAPPING T_0

At first, let the special mapping $T_0: \boldsymbol{R}^{n+m} \to \boldsymbol{R}^{n+m}$ be defined as

$$T_0(z) := \begin{bmatrix} \nabla_x L(x, u) \\ \phi^1(x, u) \\ \vdots \\ \phi^m(x, u) \end{bmatrix},$$

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where $\phi^i(x, u) := \phi(g^i(x), u^i)$ $(i \in I)$ and $z := (x, u)$. As usual, $L(x, u) := f(x) + u^T g(x)$ denotes the Lagrangian of problem (1.1). The functional $\phi : \boldsymbol{R}^2 \to \boldsymbol{R}$ is given by

$$\phi(a, b) := \sqrt{a^2 + b^2} + a - b.$$

Originally, $\phi_+(a, b)$ was constructed by Burmeister [2] as an example of a functional, which measures the distance from the cone $\boldsymbol{R}_- \times \boldsymbol{R}_+$ and possesses certain desired properties, see Fischer [7]. As easily to verify the functional ϕ is subadditive, positive homogeneous and Lipschitz-continuous. Moreover, it holds

$$\phi(a, b) = 0 \quad \text{if and only if} \quad a \leqq 0,\ b \geqq 0,\ ab = 0.$$

Thus, the equivalence of the Kuhn–Tucker conditions (1.2) to the nonlinear system $T_0(z) = 0$ follows immediately. Moreover, the mapping T_0 is locally Lipschitz-continuous, but it is Fréchet-differentiable only for points

$$z \in Z^+ := \{z \in \boldsymbol{R}^{n+m} \mid |g^i(x)| + |u^i| > 0,\ i \in I\}. \tag{2.1}$$

Therefore, the following generalized Jacobian in the sense of Clarke is employed.

Definition: Let the function $F : \boldsymbol{R}^p \to \boldsymbol{R}^q$ be Lipschitz-continuous in a neighbourhood of $z \in \boldsymbol{R}^p$, i.e. there exist positive numbers ϵ and L such that, for all $z', z'' \in B(z, \epsilon) := \{w \in \boldsymbol{R}^p \mid \|w - z\| \leqq \epsilon\}$,

$$\|F(z') - F(z'')\| \leqq L\, \|z' - z''\|.$$

Then, the convex hull of

$$\left\{D \mid \exists\{z_\nu\} : \lim_{\nu \to \infty} z_\nu = z,\ \exists\{\nabla F(z_\nu)\},\ \lim_{\nu \to \infty} \nabla F(z_\nu) = D^T\right\}$$

is called Clarke's Jacobian of F at z and is denoted by $\partial F(z)$.

This definition is taken from chapter 2.6 in [3]. Some basic properties of $\partial F(z)$ can also be found there. For instance, $\partial F(z)$ (and especially $\partial T_0(z)$) are nonempty and compact for all $z \in \boldsymbol{R}^p$. To get more information on $\partial T_0(z)$ let us at first consider the Jacobian of T_0 at a point $z \in Z^+$.

$$\nabla T_0(z) = \begin{pmatrix} \nabla^2_{xx} L(x, u) & A(x) \\ \operatorname{diag}(\gamma^i(z)) A(x)^T & \operatorname{diag}(\mu^i(z)) \end{pmatrix}^T,$$

where

$$\begin{aligned} A(x) &:= (\nabla g^1(x), \ldots, \nabla g^m(x)), \\ \gamma^i(z) &:= g^i(x)\{(g^i(x)^2 + (u^i)^2\}^{-1/2} + 1 \qquad (i \in I) \\ \mu^i(z) &:= u^i\{g^i(x)^2 + (u^i)^2\}^{-1/2} - 1 \qquad (i \in I). \end{aligned}$$

Obviously, it holds

$$\begin{aligned} \gamma^i(z) = 0 \quad &\text{if and only if} \quad g^i(x) < 0 \text{ and } u^i = 0 \qquad (i \in I), \\ \mu^i(z) = 0 \quad &\text{if and only if} \quad g^i(x) = 0 \text{ and } u^i > 0 \qquad (i \in I). \end{aligned} \tag{2.2}$$

A straightforward calculation shows further

$$\gamma^i(z) \in [0, 2],\ \mu^i(z) \in [-2, 0], \qquad (\gamma^i(z))^2 + (\mu^i(z))^2 \geqq 3 - 2\sqrt{2} \qquad (i \in I). \tag{2.3}$$

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With regard to the structure of $\nabla T_0(z)$ for $z \in Z^+$, each element of Clarke's Jacobian $\partial T_0(z)$ can be written as

$$G(z, \gamma, \mu) = \begin{pmatrix} \nabla^2_{xx} L(x, u) & A(x) \\ \operatorname{diag}(\gamma^i) A(x)^T & \operatorname{diag}(\mu^i) \end{pmatrix}, \tag{2.4}$$

where the vectors $\gamma, \mu \in \boldsymbol{R}^m$ are chosen such that $G(z, \gamma, \mu) \in \partial T_0(z)$. Owing to (2.3), $G(z, \gamma, \mu) \in \partial T_0(z)$ implies

$$\gamma^i \in [0, 2],\ \mu^i \in [-2, 0], \qquad (\gamma^i)^2 + (\mu^i)^2 \geqq 3 - 2\sqrt{2} > 0 \qquad (i \in I). \tag{2.5}$$

but not reversed. In order to describe a subset of $\partial T_0(z)$ precisely the index set

$$I^0(z) := \{i \in I \mid g^i(x) = u^i = 0\}$$

is introduced. Now, let us consider sequences $\{z_\nu\} \subset \boldsymbol{R}^{n+m}$ converging to $z \in \boldsymbol{R}^{n+m} \backslash Z^+$, such that

$$x_\nu = x \quad (\nu \in N) \quad \text{and} \quad u^i_\nu u^i_{\nu+1} > 0 \qquad (i \in I^0(z),\ \nu \in N)$$

are satisfied. Then, we have

$$\lim_{\nu \to \infty} \gamma^i(z_\nu) = 1 \quad \text{and} \quad \lim_{\nu \to \infty} \mu^i(z_\nu) \in \{-2, 0\} \qquad (i \in I^0(z)).$$

Therefore, taking into account that $\partial T_0(z)$ is the convex hull of certain limits, we obtain

$$\partial T_0(z) \supset \left\{ G(z, \gamma, \mu) \;\middle|\; \begin{array}{lll} \gamma^i = 1, & \mu^i \in [-2, 0] & \text{for } i \in I^0(z) \\ \gamma^i = \gamma^i(z), & \mu^i = \mu^i(z) & \text{for } i \in I \backslash I^0(z) \end{array} \right\} \neq \emptyset.$$

3. THE ALGORITHM

In the last years the application of Newton's method to equations $F(z) = 0$ with nondifferentiable functions F has been investigated by several authors [15, 16, 24, 18, 17]. Different concepts for replacing the classical linear subproblems of Newton's method by suitable other, possibly nonlinear, problems were presented. Related to this, generalized derivatives turn out to be essential tool.

In the paper of Kojima/Shindo [15] concerning an extension of Newton's method to piecewise continuously differentiable functions F these derivatives are specially selected linear mappings.

Pang [18] employs so-called *B-derivatives* [23], which are required to be positive homogeneous but not necessarily additive. Consequently, the subproblems of Newton's method can become nonlinear. For showing a quadratic convergence rate Pang especially assumes the Fréchet-differentiability of the function F at the solution. The B-differentiable mapping T_0 satisfies this assumption only if $z^* \in Z^+$, i.e. under the strict complementary slackness condition (1.4). Moreover, for an iterate belonging to $\boldsymbol{R}^{n+m} \backslash Z^+$ the Newton method [18] would generate a nonlinear subproblem.

Robinson [24] introduced the concept of *point-based approximations* for treating a class of nonsmooth equations. In particular, the function F may not have Fréchet derivatives at the solution. The linearization of F at a *point* $\bar{z}$ used in the conventional Newton method is replaced by an *approximation* $\bar{F}(\bar{z}, \cdot)$, such

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that $\|\bar{F}(\bar{z}, z) - F(z)\|$ is sufficiently precise in some sense for all z and all $\bar{z}$ in a neighbourhood of the solution. A Kantorovich-type theorem on the R-quadratic convergence can be proved. In our case, i.e. for $F = T_0$, such a point-based approximation would necessarily generate nonlinear subproblems if the iteration point is near by a solution $z^* \notin Z^+$.

Instead of the classical linearization of F the concept of Kummer [16, 17] uses approximations of F given by a multifunction. In order to show local superlinear convergence they are especially assumed to be sufficiently precise along rays beginning at the solution. Contrary to [18] and [24], this approach allows us to approximate $T_0(z) = 0$ by *linear equations* in such a way that local (Q-quadratic) convergence can be proved also without strict complementarity.

The algorithm we are going to state fits in the concept of Kummer. Additionally, a possibility to perturb the generated iterates is proposed.

Algorithm $\mathscr{A}$

S0: Choose $z_0 \in \boldsymbol{R}^{n+m}$ and $eps > 0$. Set $k := 0$.

S1: Choose $\hat{z}_k \in \boldsymbol{R}^{n+m}$ such that $\|\hat{z}_k - z_k\| \leqq \min(eps, \|T_0(z_k)\|)$.
Choose $G_k \in \partial T_0(\hat{z}_k)$ and compute z_{k+1} as solution of

$$T_0(\hat{z}_k) + G_k(z - \hat{z}_k) = 0.$$

S2: If $T_0(z_{k+1}) = 0$ then stop.
Otherwise, set $k := k + 1$ and go to step S1. ■

Remark 3.1: By a suitable choice of the vector $\hat{z}_k$, e.g.

$$\hat{x}_k := x_k, \qquad \hat{u}_k^i := \begin{cases} u_k^i & \text{if } |g^i(x_k)| + |u_k^i| > \epsilon_k \\ \epsilon_k & \text{otherwise} \end{cases} \qquad (i \in I), \tag{3.1}$$

with $\epsilon_k := \min(eps, \|T_0(z_k)\|)/\sqrt{4m}$ and $\|\cdot\| := \|\cdot\|_2$, it can be ensured that $\hat{z}_k \in Z^+$, i.e. $\partial T_0(\hat{z}_k) = \{\nabla T_0(\hat{z}_k)\}$. However, also the choice $\hat{z}_k := z_k$ is possible.

Remark 3.2: A generalization of Algorithm $\mathscr{A}$ can be obtained in the following way. At first, let us define

$$\tilde{\phi}^i(x, u) := \phi(\alpha^i g^i(x), \beta^i u^i), \qquad \text{where } \alpha^i, \beta^i > 0 \qquad (i \in I).$$

Then, using the functionals $\tilde{\phi}^i$ instead of ϕ^i for the definition of T_0, we get the mapping $\tilde{T}_0$ depending on the parameter vectors $\alpha, \beta \in \boldsymbol{R}^m$. (For the differentiation of $\tilde{T}_0$ the vectors α and β are handled as vectors of constants.) Different choices of α and β are imaginable. Here, it is proposed to replace Step S1 of Algorithm $\mathscr{A}$ by

Choose $\hat{z}_k \in \boldsymbol{R}^{n+m}$ such that $\|\hat{z}_k - z_k\| \leqq \min(eps, \|T_0(z_k)\|)$
Set $\alpha^i := \max^{-1}(1, |g^i(\hat{x}_k)|)$, $\beta^i := \max^{-1}(1, |\hat{u}_k^i|)$.
Choose $\tilde{G}_k \in \partial \tilde{T}_0(\hat{z}_k)$ and compute z_{k+1} as solution of

$$\tilde{T}_0(\hat{z}_k) + \tilde{G}_k(z - \hat{z}_k) = 0.$$

We note, that this modification does not destroy the theoretical properties of Algorithm $\mathscr{A}$ shown in the next Section. From our experience, it frequently leads to practical advantages concerning the global convergence behaviour.

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4. ESSENTIAL PROPERTIES

To establish a global condition for the regularity of an element of $\partial T_0(z)$ let the matrices $A^\mu(x)$ and $H(z, \gamma, \mu)$ be defined by

$$A^\mu(x) := (\cdots \nabla g^i(x) \cdots)_{i \in I^\mu}, \qquad I^\mu := \{i \in I \mid \mu^i = 0\}$$
$$H(z, \gamma, \mu) := \nabla^2_{xx} L(x, u) - \sum_{i \in I \setminus I^\mu} \gamma^i (\mu^i)^{-1} \nabla g^i(x) \nabla g^i(x)^T.$$

Theorem 4.1: *Let $z \in \boldsymbol{R}^{n+m}$ be arbitrary but fixed. The vectors $\gamma \in \boldsymbol{R}^m$ and $\mu \in \boldsymbol{R}^m$ be such that $G(z, \gamma, \mu) \in \partial T_0(z)$. Then, $G^{-1}(z, \gamma, \mu)$ exists if and only if the matrix*

$$P(z, \gamma, \mu) := \begin{pmatrix} H(z, \gamma, \mu) & A^\mu(x) \\ A^\mu(x)^T & 0 \end{pmatrix}$$

is regular.

Proof: For an arbitrary but fixed $G(z, \gamma, \mu) \in \partial T_0(z)$, let us consider the homogeneous system

$$G(z, \gamma, \mu) \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = 0 \qquad h_1 \in \boldsymbol{R}^n, h_2 \in \boldsymbol{R}^m$$

In view of (2.5), this system can be transformed equivalently into

$$P(z, \gamma, \mu) \begin{pmatrix} h_1 \\ h_2^\mu \end{pmatrix} = 0, \qquad h_2^i = -\gamma^i (\mu^i)^{-1} \nabla g^i(x)^T h_1 \qquad (i \in I \setminus I^\mu)$$

with $h_2^\mu := (\cdots h_2^i \cdots)^T_{i \in I^\mu}$. The regularity of $P(z, \gamma, \mu)$ then implies $h_1 = 0$ and $h_2 = 0$, i.e. $G^{-1}(z, \gamma, \mu)$ exists and vice versa. ■

Remark 4.1: As an immediate consequence of Theorem 4.1, the regularity of $G(z, \gamma, \mu) \in \partial T_0(z)$ requires the linear independence of the vectors $\nabla g^i(x)$ $(i \in I^\mu)$.

Remark 4.2: From (2.2), it follows for the mapping T_0 that

$$I^+(z) \subseteq I^\mu \subseteq I(x).$$

According to Remark 4.1, at most the gradients of the active constraints must be linearly independent.

To proceed we make use of the following standard assumptions, where $z^* = (x^*, u^*)$ denotes a Kuhn–Tucker point of problem (1.1).

Assumptions

(A1) The mappings $\nabla^2 f, \nabla^2 g^1, \ldots, \nabla^2 g^m$ are Lipschitz-continuous in a neighbourhood of x^*.

(A2) The vectors $\nabla g^i(x^*)$ $(i \in I(x^*))$ are linearly independent.

(A3) The strong second order sufficiency condition is satisfied, i.e.

$$s^T \nabla^2_{xx} L(x^*, u^*) s > 0 \quad \text{for} \quad s \neq 0 \quad \text{with} \quad \nabla g^i(x^*)^T s = 0 \qquad (i \in I^+(x^*, u^*)).$$

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Lemma 4.2: *Let the Assumptions (A2) and (A3) be satisfied. Then, there are positive numbers δ and M such that, for all $G \in \partial T_0(z)$,*

$$G \text{ is regular and } \|G^{-1}\| \leqq M,$$

where $z \in B(z^*, \delta)$.

Proof: For an arbitrary but fixed $G(z^*, \gamma, \mu) \in \partial T_0(z^*)$ we consider the matrices $A^\mu(x^*)$, $H(z^*, \gamma, \mu)$ and $P(z^*, \gamma, \mu)$. Remark 4.2 and Assumption (A2) yield the linear independence of the columns of $A^\mu(x^*)$. Since $-\gamma^i(\mu^i)^{-1}y^T \nabla g^i(x^*) \nabla g^i(x^*)^T y \geqq 0$ for all $y \in \boldsymbol{R}^n$ and all $i \in I \backslash I^\mu$, we obtain by Assumption (A3) and Remark 4.2 that

$$y^T H(z^*, \gamma, \mu) y > 0 \quad \text{for} \quad y \neq 0 \quad \text{with} \quad A^\mu(x^*) y = 0.$$

As a well-known consequence of these facts, the regularity of $P(z^*, \gamma, \mu)$ follows. Therefore, in view of Theorem 4.1, $G(z^*, \gamma, \mu)^{-1}$ exists. The same holds for all $G \in \partial T_0(z^*)$. Owing to the compactness of $\partial T_0(z^*)$ we have $\sup\{\|G^{-1}\| \mid G \in \partial T_0(z^*)\} < \infty$. By Theorem 2.6.2c) of [3], the point-to-set mapping ∂T_0 is upper semicontinuous at z^*, since T_0 is Lipschitz-continuous. Consequently, there exist positive numbers δ and M such that

$$\sup\{\|G^{-1}\| \mid G \in \partial T_0(z), z \in B(z^*, \delta)\} \leqq M. \quad \blacksquare$$

Lemma 4.3: *There exist positive numbers ϵ and C such that, for all $\Delta^i \in \partial \phi^i(z)$,*

$$|\phi^i(z) - \Delta^i(z - z^*)| \leqq C \|z - z^*\|^2 \qquad (i \in I), \tag{4.1}$$

where $z \in B(z^*, \epsilon)$.

Proof. Setting $r^i := g^i(x) - \nabla g^i(x)^T(x - x^*)$ and using Taylor's formula, positive numbers ϵ_0 and c_0 can be found such that for all $z \in B(z^*, \epsilon_0)$,

$$|r^i| \leqq c_0 \|z - z^*\|^2 \qquad (i \in I(x^*)). \tag{4.2}$$

From now, we assume $z \in B(z^*, \epsilon_0)$.

(a) At first, let us consider the case $i \in I^0(z^*) \backslash I^0(z)$. Then, we have $\partial \phi^i(z) = \{\nabla^T \phi^i(z)\}$, since $\nabla \phi^i(z)$ exists and is continuous. Consequently, we obtain

$$\Delta^i(z - z^*) = \nabla \phi^i(z)^T(z - z^*) = \gamma^i(z) \nabla g^T(x)(x - x^*) + \mu^i(z)(u^i - u^{*i}).$$

Because of $u^{*i} = 0$, it follows

$$\Delta^i(z - z^*) = \gamma^i(z) g^i(x) - \gamma^i(z) r^i + \mu^i(z) u^i = \phi^i(z) - \gamma^i(z) r^i.$$

With regard to (4.2) and owing to $\gamma^i(z) \in [0, 2]$, this yields

$$|\phi^i(z) - \Delta^i(z - z^*)| \leqq 2c_0 \|z - z^*\|^2.$$

b) Now, let $i \in I^0(z^*) \cap I^0(z)$ be satisfied. From a chain rule for Clarke's Jacobian ([3], Theorem 2.6.6), we obtain

$$\partial \phi^i(z) \subseteq \partial \phi(g^i(x), u^i) \begin{pmatrix} \nabla g^i(x)^T & 0, \ldots, 0 \\ 0, \ldots, 0 & e_i^T \end{pmatrix},$$

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where $e_i \in \boldsymbol{R}^m$ denotes the ith unit vector. By (2.5) and $u^i = u^{*i} = 0$, this leads to

$$\partial\phi^i(z)(z - z^*) \subset [0, 2] \times [-2, 0]\begin{pmatrix} \nabla g^i(x)^T(x - x^*) \\ 0 \end{pmatrix}. \tag{4.3}$$

Since $g^i(x) = g^i(x^*) = 0$, relation (4.2) yields

$$|\nabla g^i(x)^T(x - x^*)| \leqq c_0 \, \|z - z^*\|^2.$$

Using this, from (4.3) and $\phi^i(z) = 0$ it follows that

$$|\phi^i(z) - \Delta^i(z - z^*)| \leqq 2c_0 \, \|z - z^*\|^2.$$

c) Finally, let be $i \in I \backslash I^0(z^*)$. Then, for $\epsilon_i > 0$ sufficiently small, the functional ϕ^i is twice differentiable for all $z \in B(z^*, \epsilon_i)$. Since $\phi^i(z^*) = 0$, Taylor's formula implies relation (4.1) with $\epsilon = \epsilon_i$ and for a suitable $C = c_i > 0$.

Setting $\epsilon := \min \epsilon_i$ and $C := 2 \max c_i$ for $i \in \{0\} \cup I \backslash I^0(z^*)$, the proof is complete. ■

Lemma 4.4: *Let the Assumptions (A1), (A2) and (A3) be satisfied. Then, there exist positive numbers $\bar{\epsilon}$ and $\bar{C}$ such that, for all $G \in \partial T_0(z)$,*

$$\|T_0(z) - G(z - z^*)\| \leqq \bar{C} \, \|z - z^*\|^2,$$

where $z \in B(z^*, \bar{\epsilon})$.

Proof: By Theorem 2.6.2e) of [3] and in view of Assumption (A1), we know that

$$\partial T_0(z) \subseteq \begin{pmatrix} \{\nabla^2_{xx} L(x, u) - A(x)\} \\ \partial\phi^1(z) \\ \vdots \\ \partial\phi^m(z) \end{pmatrix}. \tag{4.4}$$

Because of Assumption (A1), Taylor's formula yields

$$\|\nabla_x L(x, u) - \nabla^2_{xx} L(x, u)(x - x^*) - A(x)(u - u^*)\| \leqq \bar{c} \, \|z - z^*\|^2$$

for $z \in B(z^*, \bar{\epsilon})$ with suitable positive numbers $\bar{\epsilon}$ and $\bar{c}$. Therefore, together with Lemma 4.3 the assertion of Lemma 4.4 is obvious, since $\bar{\epsilon} \leqq \epsilon$ can be assumed. ■

Theorem 4.5: *Let the Assumptions (A1), (A2) and (A3) be satisfied. Further, let be $z_0 \in B(z^*, \bar{\epsilon})$ with $\bar{\epsilon} > 0$ sufficiently small. Then, Algorithm $\mathcal{A}$ generates a well defined sequence $\{z_k\} \subset B(z^*, \bar{\epsilon})$. Moreover, if this sequence is infinite it converges to z^* and there is $\bar{C} > 0$ such that, for all $k \in N$,*

$$\|z_{k+1} - z^*\| \leqq \bar{C} \, \|z_k - z^*\|^2.$$

Proof: Let $z \in B(z^*, \bar{\epsilon})$, $\hat{z} \in B(z, \|T_0(z)\|)$ and $G \in \partial T_0(\hat{z})$ be arbitarily chosen. By the local Lipschitz-continuity of T_0 there exists a number $\bar{L}$ such that

$$\|T_0(z)\| = \|T_0(z) - T_0(z^*)\| \leqq \bar{L} \, \|z - z^*\|. \tag{4.5}$$

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This implies

$$\|\hat{z} - z^*\| \leqq \|\hat{z} - z\| + \|z - z^*\| \leqq \|T_0(z)\| + \bar{\epsilon} \leqq \bar{\epsilon}(1 + \bar{L}). \tag{4.6}$$

Thus, for $\bar{\epsilon} > 0$ sufficiently small, Lemma 4.2, ensures that G is regular with $\|G^{-1}\| \leqq M$. For

$$\tilde{z} := \hat{z} - G^{-1}T_0(\hat{z}) \tag{4.7}$$

we obtain

$$\begin{aligned} \|\tilde{z} - z^*\| &= \|\hat{z} - G^{-1}T_0(\hat{z}) - z^*\| \leqq \|G^{-1}\| \, \|T_0(\hat{z}) - G(\hat{z} - z^*)\| \\ \|\tilde{z} - z^*\| &\leqq M \, \|T_0(\hat{z}) - G(\hat{z} - z^*)\|. \end{aligned} \tag{4.8}$$

Since $\bar{\epsilon}(1 + \bar{L}) \leqq \bar{\epsilon}$ and $M\bar{C}\bar{\epsilon}(1 + \bar{L})^2 \leqq 1$ can be assumed, Lemma 4.4 is applicable to (4.8), and with (4.6) we get

$$\|\tilde{z} - z^*\| \leqq M\bar{C} \, \|\hat{z} - z^*\|^2 \quad \text{and} \quad \tilde{z} \in B(z^*, \bar{\epsilon}).$$

This inequality, the first one of (4.6) and $\|\hat{z} - z\| \leqq \|T_0(z)\|$ with regard to relation (4.5) yield

$$\|\tilde{z} - z^*\| \leqq M\bar{C}(1 + \bar{L})^2 \, \|z - z^*\|^2.$$

Setting $z := z_k$, $\hat{z} := \hat{z}_k$, $G := G_k$ and comparing (4.7) with the linear system in Step S1 of Algorithm $\mathscr{A}$ we see that $z_{k+1} = \tilde{z}$. In view of $\tilde{z} \in B(z^*, \bar{\epsilon})$, Theorem 4.5 follows by induction, where $M\bar{C}(1 + \bar{L})^2$ is a possible value for $\bar{C}$. ■

Remark 4.3: The proof of Theorem 4.5 makes use of a technique in Kojima/Shindo [15]. Moreover, utilizing Lemma 4.2 and Lemma 4.4, we would be able to deduce the above Theorem for the unperturbed version ($\hat{z}_k = z_k$) of Algorithm $\mathscr{A}$ from a general result of Kummer [16]. Using this approach, a Q-superlinear rate of convergence can also be proved on condition that the functionals $f, g^1, \ldots, g^m$ are only one time differentiable with Lipschitz-continuous derivatives, if additional assumptions, in particular concerning $\partial\nabla_x L(x^*, u^*)$, are satisfied.

5. COMPARISON WITH EXISTING METHODS

Based on Algorithm $\mathscr{A}$ some examples of NOM given by the different mappings T_1, T_2 and T_3 are considered in the following.

Example 5.1:

$$T_1(z) := \begin{bmatrix} \nabla_x L(x, \tilde{v}) \\ g(x) + \tilde{y} \\ y^1 v^1 \\ \vdots \\ y^m v^m \end{bmatrix}, \qquad \begin{aligned} &z := (x, v, y) \in \boldsymbol{R}^n \times \boldsymbol{R}^m \times \boldsymbol{R}^m, \qquad p := n + 2m \\ &\tilde{v} := \tfrac{1}{2}((v^1)^2, \ldots, (v^m)^2)^T \\ &\tilde{y} := \tfrac{1}{2}((y^1)^2, \ldots, (y^m)^2)^T \end{aligned}$$

This mapping was introduced by Abadie [1] and Kleinmichel/Schönefeld [11] independently. The last two authors widely investigated NOM (cf. [11, 12, 13,

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26]). The Jacobian of T_1 exists everywhere and is given by

$$\nabla T_1(z) = \begin{pmatrix} \nabla^2_{xx}L(x, \tilde{v}) & A(x)\,\mathrm{diag}(v^i) & 0 \\ A(x)^T & 0 & \mathrm{diag}(y^i) \\ 0 & \mathrm{diag}(y^i) & \mathrm{diag}(v^i) \end{pmatrix}^T.$$

Remark 5.1: In order to ensure a local superlinear rate of convergence of Algorithm $\mathscr{A}$ using the mapping T_1 instead of T_0 an assertion analogous with Lemma 4.2 has to be proved, i.e. the Jacobian $\nabla T_1^T(z^*)$ must be regular (cf. [11, 26]). Evidently, this is only possible if the strict complementary slackness condition (1.4) is fulfilled.

A drawback of the mapping T_1 is the fact that $v_k^i = 0$ implies $v_{k+1}^i = 0$, and so on, even if $(v^{*i})^2 = 2u^{*i} > 0$ holds. The analogous situation appears if $y_k^i = 0$, (see [13]).

Remark 5.2: Let the weak condition $|v^i| + |y^i| > 0$ be satisfied for all $i \in I$. A consideration of the homogeneous linear system $(h_1^T, h_2^T, h_3^T)\nabla T_1(z) = 0$ with $h_1 \in \boldsymbol{R}^n$, $h_2, h_3 \in \boldsymbol{R}^m$ yields

$$\begin{aligned} h_2^i &= v^i(y^i)^{-2}\nabla g^i(x)^T h_1 && \text{if } y^i \neq 0 \\ h_3^i &= -(y^i)^{-1}\nabla g^i(x)^T h_1 && \text{if } y^i \neq 0 \\ h_3^i &= 0 && \text{if } y^i = 0 \end{aligned}$$

Ssetting $\gamma^i := (v^i)^2$ and $\mu^i := -(y^i)^2$ for all $i \in I$ we can use the definitions of I^μ, $A^\mu(x)$, h_2^μ in Section 4 and obtain the reduced system

$$\left(\nabla^2_{xx}L(x, \tilde{v}) - \sum_{i \in I \setminus I^\mu} \gamma^i(\mu^i)^{-1}\nabla g^i(x)\nabla g^i(x)^T\right)h_1 + A^\mu(x)h_2^\mu = 0$$

$$A^\mu(x)^T h_1 = 0$$

Replacing the vector u in the definition of $H(z, \gamma, \mu)$ by $\tilde{v}$ we see that the last system is uniquely solvable if and only if $P(z, \gamma, \mu)$ in Theorem 4.1 is regular. With $G(z) := \nabla T_1(z)$ instead of $G(z, \gamma, \mu) \in \partial T_0(z)$, Remark 4.1 holds for the mapping T_1 analogously.

Example 5.2:

$$T_2(z) := \begin{bmatrix} \nabla_x L(x, u) \\ (u + g(x))_+ - u \end{bmatrix}, \qquad z := (x, u) \in \boldsymbol{R}^n \times \boldsymbol{R}^m, \qquad p := n + m$$

This mapping was proposed by Wierzbicki [30]. Provided that $u^i + g^i(x) \neq 0$ $(i \in I)$, we have

$$\nabla T_2(z) = \begin{pmatrix} \nabla^2_{xx}L(x, u) & A(x) \\ \mathrm{diag}(\gamma^i(z))A(x)^T & \mathrm{diag}(\mu^i(z)) \end{pmatrix}^T,$$

where

$$\gamma^i(z) := \begin{cases} 1 & \text{if } u^i + g^i(x) > 0 \\ 0 & \text{if } u^i + g^i(x) < 0 \end{cases}, \qquad \mu^i(z) := \gamma^i(z) - 1 \qquad (i \in I).$$

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Otherwise, the Jacobian does not exist. Then, each element of Clarke's Jacobian $\partial T_2(z)$ has the structure (2.4) and can be denoted by $G(z, \gamma, \mu)$ with suitable vectors $\gamma, \mu \in \boldsymbol{R}^m$.

Remark 5.3: In analogy with the mapping T_0, it is possible to show that Theorem 4.1 and Remark 4.1 also hold for T_2, where (2.5) corresponds to

$$\gamma^i \in [0, 1], \qquad \mu^i \in [-1, 0], \qquad (\gamma^i)^2 + (\mu^i)^2 \geqq \tfrac{1}{2} \qquad (i \in I).$$

However, for the mapping T_2 we obtain the inclusions

$$\{i \in I \mid u^i + g^i(x) > 0\} \subseteq I^\mu \subseteq \{i \in I \mid u^i + g^i(x) \geqq 0\}.$$

Therefore, the linear independence condition on the vectors $\nabla g^i(x)$ $(i \in I^\mu)$ derived in Remark 4.1 is a strong one. This can be demonstrated by the first test problem given in Section 6. With regard to the constraints of this problem, let us consider the case

$$z \in \{(x, u) \in \boldsymbol{R}^2 \times \boldsymbol{R}^3 \mid |x_1| \leqq 1, |x_2| \leqq 1, u_1 > -8, u_2 > 0, u_3 > 0\}.$$

Then, we have $I^\mu = I$, i.e. the necessary linear independence condition on the vectors $\nabla g^i(x)$ $(i \in I^\mu)$ cannot be satisfied.

Example 5.3:

$$T_3(z) := \begin{bmatrix} \nabla_x L(x, w_+) \\ g + (-w)_+ \end{bmatrix}, \qquad z := (x, w) \in \boldsymbol{R}^n \times \boldsymbol{R}^m, \qquad p := n + m$$

This mapping is due to Kojima/Hirabayashi [14]. If $w^i \neq 0$ $(i \in I)$, we have

$$\nabla T_3(z) = \begin{pmatrix} \nabla^2_{xx} L(x, w_+) & A(x)\,\mathrm{diag}(\mu^i(z) + 1) \\ \mathrm{diag}(\gamma^i(z)) A(x)^T & \mathrm{diag}(\mu^i(z)) \end{pmatrix}^T,$$

where

$$\mu^i(z) := \begin{cases} -1 & \text{if } w^i < 0 \\ 0 & \text{if } w^i > 0 \end{cases} \quad \text{and} \quad \gamma^i(z) := 1 \qquad (i \in I).$$

Remark 5.4: If the matrix $H(z, \gamma, \mu)$ in the definition of $P(z, \gamma, \mu)$ is replaced by $\nabla^2_{xx} L(x, w_+)$ and $w^i \neq 0$ $(i \in I)$ holds, then, similarly to the proof of Theorem 4.1, it can be shown that $\nabla T_3(z)$ is regular if and only if $P(z, \gamma, \mu)$ has this property. Moreover, Remark 4.1 is also valid for the mapping T_3. (Note that the structure of the matrices $G(z, \gamma, \mu) \in \partial T_3(z)$ slightly differs from the structure (2.4) related to the mapping T_0.) Unfortunately, the linear independence condition on the vectors $\nabla g^i(x)$ $(i \in I^\mu)$ coming from Remark 4.1 is a strong one again, since

$$\{i \in I \mid w^i > 0\} \subseteq I^\mu \subseteq \{i \in I \mid w^i \geqq 0\}.$$

Applied to Problem 6.1, that means that the matrix G_k of the linear system in Step S1 of Algorithm $\mathscr{A}$ is singular if $w^i_k > 0$ for $i = 1, 2, 3$.

Remark 5.5: Purtov [20] describes a class of mappings $T: \boldsymbol{R}^{n+m} \to \boldsymbol{R}^{n+m}$ satisfying stronger differentiability conditions than T_0, T_2 or T_3 and suggests a perturbation of the mapping T to avoid the singularity of $\nabla T(z)$ far from a solution z^*. However, this technique is based on the strict complementary slackness condition (1.4).

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Remark 5.6: The use of the mapping T_2 or T_3 implies a strong condition concerning the linear independence of certain vectors $\nabla g^i(x)$ (cf. Remarks 5.3 and 5.4). In contrast to this, the corresponding requirements for the mapping T_0 or T_1 seem to be reasonable, (cf. Remarks 4.2 and 5.2). However, for the mapping T_1 the strict complementary slackness condition (1.4) is necessary to ensure the regularity of $\nabla T_1(z^*)$ (cf. Remark 5.1). On the other hand, Algorithm $\mathscr{A}$ converges locally superlinearly without assuming the condition (1.4), if one of the mappings T_0, T_2 or T_3 is employed. (For the special piecewise-continuously differentiable mappings T_2 and T_3 this fact follows from [15].)

Therefore, with regard to the properties i) and ii) of Section 1 the mapping T_0 is favoured over against T_1, T_2 and T_3. By analogous arguments, the same can be said if we compare the Algorithm $\mathscr{A}$ using T_0 with other possible Newton-type Optimization Methods based on the literature (cf. [1, 6, 13, 20, 26]).

6. NUMERICAL EXPERIENCE

By means of two test problems we will report on the practical behaviour of Algorithm $\mathscr{A}$ coded in Turbo–Pascal (double precision arithmetic). Instead of T_0 the modified mapping $\tilde{T}_0$ according to Remark 3.2 is used. From our experience, the region of convergence for $\tilde{T}_0$ seems to be more extended.

For the sake of comparison, results are also given in the case that the mapping T_0 in Algorithm $\mathscr{A}$ is replaced by T_2 or T_1. (Results for T_3 are not reported. They are qualitatively similar to the ones for T_2.) It has to be noted that the test of Algorithm $\mathscr{A}$ with T_1 (and also T_3) cannot directly be compared with the calculations using T_0 and T_2. In particular, different starting vectors are required. For the mapping T_1 the vector $z_0 = (x_0, v_0, y_0)$ satisfying $(v_0^i)^2 := 2u_0^i$, $v_0^i > 0$, $y_0^i := 1$ $(i \in I)$ was used. Consequently, to define v_0 only such starting vectors (x_0, u_0) of Table 1 or 2 with $u_0^i > 0$ $(i \in I)$ are suitable.

The implementation of Algorithm $\mathscr{A}$ includes the perturbation rule (3.1) with $eps := 10^{-5}$ so that $G_k \in \partial \tilde{T}_0(z_k) = \{\nabla \tilde{T}_0(z_k)\}$. If for the mapping T_2 the Jacobian $\nabla T_2(z_k)$ does not exist, then Algorithm $\mathscr{A}$ chooses $G_k = G(z_k, \gamma_k, \mu_k) \in \partial T_2(z_k)$ such that $\gamma_k^i = 0$ and $\mu_k^i = -1$ for all $i \in I$ with $g^i(x_k) + u_k^i = 0$, (see Example 5.2).

Table 1 Results for Problem 5.1

x_0^T	u_0^T	$\tilde{T}_0$		T_2		T_1
		$\tau = 1$	$\tau = \frac{2}{3}$	$\tau = 1$	$\tau = \frac{2}{3}$	$\tau = 1$
(1.5, 2.5)	(1, 1, 1)	7	6	4	4	7
(0, 0)	(1, 1, 1)	10	17	s	s	15
(−10, −10)	(1, 1, 1)	12	21	n	n	19
(−100, −100)	(1, 1, 1)	n	39	n	n	n
(1, 1)	(1, 1, 1)	19	10	s	s	9
(3, 3)	(3, 3, 3)	8	7	s	s	8
(10, 10)	(1, 1, 1)	10	9	n	n	12
(1, 1)	(0, 0, 1)	11	n	n	n	—
(10, 10)	(−1, −1, −1)	n	s	n	n	—
(0, 0)	(0, 0, 0)	11	9	s	s	—
(0, 0)	(−1, −1, −1)	11	11	s	s	—

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The Tables 1 and 2 contain the number k of iterations required to achieve the accuracy $\|z_k - z^*\|_2 < 10^{-10}$ for various given starting points $z_0 = (x_0, u_0)$. By the characters "n" and "s" the cases "no convergence" and "G_k is singular for some $k \leqq 100$" are denoted. In the first case the sequence $\{z_k\}$ either tends to infinity or the iteration number $k = 100$ is exceeded. The first starting point of Table 1 and Table 2 respectively was used for Table 3. Besides other numerical reasons the possible accuracy is restricted there, since all first and second order derivatives of f and $g^1, \ldots, g^m$ were discretized.

Problem 6.1:

$$f(x) := \tau x_1^3 + x_2^2$$

$$g_1(x) := 10 - x_1^2 - x_2^2, \qquad g_2(x) := 1 - x_1, \qquad g_3(x) := 1 - x_2$$

$$\tau = 1\colon\ x^* = (1, 3)^T\ u^* = (1, 1, 0)^T\ I(x^*) = \{1, 2\}$$

$$\tau = 2/3\colon\ x^* = (1, 3)^T\ u^* = (1, 0, 0)^T\ I(x^*) = \{1, 2\}$$

For $\tau = 1$, this problem was suggested by Tone [29]. If $\tau = 2/3$, the strict complementary slackness condition (1.4) is not fulfilled. Moreover, the test problem contains 3 constraints but only 2 variables. According to the Remarks 5.3 and 5.4, the use of the mapping T_2 or T_3 can lead to inconsistent subproblems for many starting or iteration points, cf. case "s" in Table 1. The case "n" in the column "T_2" of the same table mostly appears since the iteration cycles.

Problem 6.2:

$$\begin{aligned} f(x) &:= (x_1 - 10)^2 + 5(x_2 - 12)^2 + x_3^4 + 3(x_4 - 11)^2 \\ &\quad + 10x_5^6 + 7x_6^2 + x_7^4 - 4x_6x_7 - 10x_6 - 8x_7 \\ g^1(x) &:= 2x_1^2 + 3x_2^4 + x_3 + 4x_4^2 + 5x_5 - 127 \\ g^2(x) &:= 7x_1 + 3x_2 + 10x_3^2 + x_4 - x_5 - 282 + \lambda \\ g^3(x) &:= 23x_1 + x_2^2 + 6x_6^2 - 8x_7 - 196 \end{aligned}$$

Table 2 Results for Problem 5.2

x_0^T	u_0^T	$\bar{T}_0$		T_2		T_1
		$\lambda = 0$	$\lambda = \bar{\lambda}$	$\lambda = 0$	$\lambda = \bar{\lambda}$	$\lambda = 0$
(2, 2, 0, 4, −1, 1, 2)	(1, 0.1, 0.1, 1)	9	8	n	n	12
(2.5, 2, −0.5, 4.5, −0.5, 1, 1.5)	(1, 0.1, 0.1, 0.5)	6	6	5	5	15
(2, 2, 0, 4, −1, 1, 2)	(1, . . . , 1)	11	29	32	n	18
(1, 2, 0, 4, 0, 1, 1)	(1, . . . , 1)	23	21	n	n	30
(1, 2, 0, 4, 0, 1, 1)	(10, . . . , 10)	19	12	20	n	30
(1, . . . , 1)	(1, 0.1, 0.1, 1)	37	23	n	26	28
(1, . . . , 1)	(1, . . . , 1)	29	23	29	60	16
(10, . . . , 10)	(1, 0.1, 0.1, 1)	17	20	39	21	17
(10, . . . , 10)	(1, . . . , 1)	30	39	39	21	17
(1, . . . , 1)	(0, 1, 1, 0)	16	21	60	58	—
(0, . . . , 0)	(1, . . . , 1)	35	31	99	n	40
(0, . . . , 0)	(−1, . . . , −1)	45	n	n	n	—
(0, . . . , 0)	(0, . . . , 0)	31	31	n	n	—
(−10, . . . , −10)	(1, . . . , 1)	30	27	n	n	17

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Table 3 Local behavior of Algorithm $\mathscr{A}$ given by $\|z_k - z^*\|_2$

k	Problem 5.1		Problem 5.2	
	$\tau = 1$	$\tau = \frac{2}{3}$	$\lambda = 0$	$\lambda = \bar{\lambda}$
0	1.2 E 0	16 E 0	1.1 E 0	1.1 E 0
1	1.8 E 0	1.5 E 0	5.6 E-1	5.6 E-1
2	9.7 E-1	2.3 E-1	1.0 E 0	8.7 E-1
3	5.7 E-1	2.1 E-2	5.1 E-1	3.3 E-1
4	1.6 E-1	2.1 E-4	1.9 E-1	8.2 E-2
5	5.4 E-3	2.2 E-8	4.0 E-2	8.7 E-3
6	2.2 E-6	1.0 E-12	2.4 E-3	3.1 E-4
7	1.1 E-12		2.1 E-5	1.2 E-6
8			6.9 E-9	6.7 E-11

$$g^4(x) := 4x_1^2 + x_2^2 - 3x_1x_2 + 2x_3^2 + 5x_6 - 11x_7$$

$$x^* = (2.33\ldots, 1.95\ldots, -0.47\ldots, 4.36\ldots, -0.62\ldots, 1.03\ldots, 1.59\ldots)^T$$

$$u^* = (1.13\ldots, 0, 0, 0.36\ldots)^T$$

$$\lambda = 0\colon\ I(x^*) = \{1, 4\} \qquad \lambda = \bar{\lambda} := 252.56\ldots\colon\ I(x^*) = \{1, 2, 4\}$$

For $\lambda = 0$, this problem is due to Wong, (cf. [9]). The value $\bar{\lambda}$ instead of $\lambda = 0$ leads to a violation of the strict complementary slackness condition (1.4). Summarizing our calculations and those ones of Kleinmichel/Schönefeld [11, 12] it turns out that the use of $\tilde{T}_0$ or T_1 leads to similarly good results if the original problem (1.1) satisfies the strict complementary slackness condition (1.4). Otherwise, near a Kuhn–Tucker point the mapping T_1 cannot be employed successfully, cf. Remark 5.1. The mappings $\tilde{T}_0$, T_2 (and T_3) avoid this disadvantage. However, $\tilde{T}_0$ shows the better global behavior.

If Algorithm $\mathscr{A}$ with the mapping $\tilde{T}_0$ (or T_0) is employed the Tables 1–3 and further experience confirm that it is not significant whether an optimization problem satisfies the strict complementary slackness condition or not. In most cases the iteration sequence $\{z_k\}$ converges to the Kuhn–Tucker point z^*, although only local convergence can be expected. On the other hand, the number of required iterations is very large for several cases. In order to improve the global convergence behavior an additional technique is desirable. A hybrid principle (cf. [26]) or a step length procedure could be used. For damping techniques in the field of generalized Newton methods we refer to [18] and [21].

Acknowledgement

I would like to thank the referees and Prof. B. Kummer for valuable remarks and for directing my attention to some of the latest related publications.

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